

Batch: **Roll No.:**
Experiment / assignment / tutorial No. 06
Grade: AA / AB / BB / BC / CC / CD /DD
Signature of the Staff In-charge with date

Title: Pulse Width Modulation using IC 555

Objectives:

1. To mount the circuit of PWM and observe the waveforms.

OUTCOME:

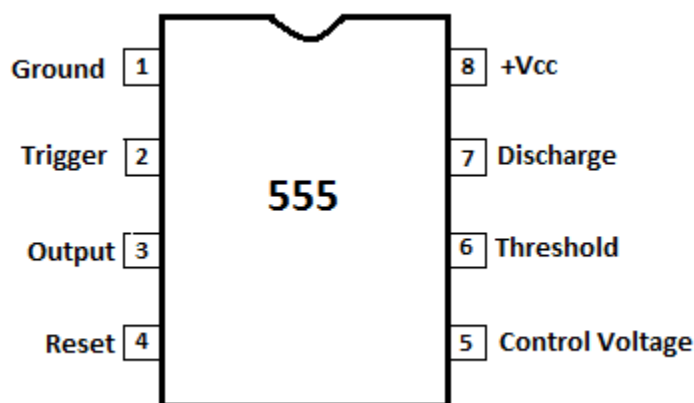
1. understand working of analog communication transmitter and receiver systems
2. compare Pulse modulation systems and multiplexing techniques.
3. understand noise effect on the performance of communication systems.

• **Procedure:**

Pulse Width Modulation:

1. Make the connection as figure.
2. Connect the supply $V_{cc} = 10V$ to the circuit.
3. Apply the sinusoidal modulating signal of amplitude 5 to 10Vpp and the frequency in the range of the 100 Hz to 500Hz.
4. Select the square wave as a carrier of amplitude is 8 to 12Vpp and frequency in the range of the 1 KHz to 2 KHz.
5. Observe the PWM output at pin no 3 of the IC555 without any DC shift of the modulating signal.
6. Observe the PWM output with DC shift of the modulating signal.
7. Plot the waveforms on the graph paper with and without shift of the modulating signal. Note the variation of the modulating and carrier frequency on the PWM Output.

- **Pin Diagram of IC 555:** (Image source: <https://www.electronicsforu.com/technology-trends/learn-electronics/555-timer-working-specifications>)



- **Draw Functional block diagram of IC 555**
- **Circuit diagram: Draw circuit diagram of PWM**

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- **Observation Table:**

Frequency of Modulating Signal	Amplitude of Modulating Signal	Frequency of Carrier Signal	Amplitude of Carrier Signal

- **Post lab activity:**

- Explain the following applications of PWM in brief:
 - Changing the Screen's Brightness By Utilizing PWM
 - Servo Control
 - Driving a buzzer

Signature of faculty in-charge